

■ Automated Aquaponics System Diagnostic Report

Generated Timestamp: 2026-06-16 11:16:55

1. Real-time Environmental Telemetry Matrix

System Parameter	Current Simulation Value	Operational Target Status
Water Temperature	26.80 °C	Optimal: 23 - 28 °C
Water pH Level	7.78	Optimal: 6.5 - 7.5
Dissolved Oxygen (O2)	8.14 mg/L	Optimal: > 5.5 mg/L
Atmospheric Humidity	63.62 %	Optimal: 40 - 80 %
Water Tank Level	74.96 %	Optimal: > 50 %
Ammonia (NH3) Load	0.16 ppm	Optimal: < 0.25 ppm
Nitrate (NO3) Level	15.00 ppm	Optimal: 10 - 40 ppm
Hydro-Pump Flow Rate	0.59 L/min	Optimal: > 1.0 L/min

2. Autonomous Root-Cause & Actionable Diagnostics

- **All Core Matrices Stable:** No biological deviations detected across the ecosystem.
- Ecosystem Report Status: The Nitrogen cycle is fully balanced. Plants absorbing nitrate ions at standard metabolic rates.

3. Fish Biomass Configuration Status

- Total Fish Count: **250 Pcs** | Net Calculated Biomass: **58.75 kg**
- Global Ecosystem Health Rating: **100/100** | Feed Efficiency Score: **70/100**